



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

15-07-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	D	A	A	B	A	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	D	A	A	C	B	C	D	B

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	D	D	C	C	B	B	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	A	B	D	D	B	B	D	B

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	D	A	D	A	C	C	D	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	B	C	B	B	B	B	A	D

HINT – SHEET

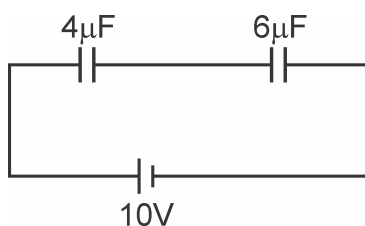
PART-1 : PHYSICS

SECTION-I

1. **Ans (B)**

$$\frac{1}{C_{eq}} = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

$$C_{eq} = \frac{12}{5} \mu F$$



$$V_2 = \frac{4}{10} \times 10 = 4V$$

2. **Ans (C)**

$w = F \times \text{displacement of point of application (x)}$

$$x = vt$$

$$v_1 = v + v = 2v$$

$$v_2 = v + \frac{v}{2} = \frac{3v}{2}$$

$$\frac{w_1}{w_2} = \frac{x_1}{x_2} = \frac{2vt}{\frac{3vt}{2}} = \frac{4}{3}$$

3. **Ans (B)**

$$I_1 = I_0 + md^2 \Rightarrow I_0 = \frac{mR^2}{4}$$

$$I = I_0 + m, d^{12} = \frac{mR^2}{4} + mR^2 = \frac{5mR^2}{4}$$

4. **Ans (D)**

$w_f = -\mu mgh$ in each case

5. **Ans (A)**

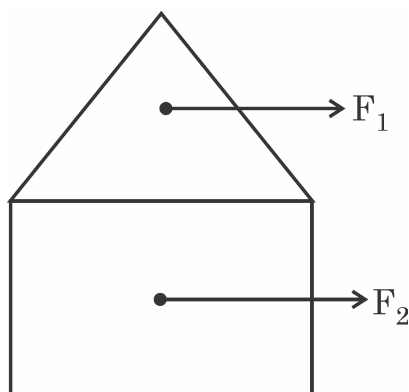
Work done = Force $\times$ Displacement of point of application.

6. **Ans (A)**

$$\vec{r}_{\text{com}} = \vec{r}_0 + \vec{v}_{\text{com}} t$$

$$\begin{aligned}\vec{r}_{\text{com}} &= \frac{2(1, 2) + 3(2, 3)}{2 + 3} \\ &+ \frac{2(1, 2) + 3(2, -1)}{5} \times 5 \\ &= \left(\frac{8}{5}, \frac{13}{5}\right) + (8, 1) = \left(\frac{48}{5}, \frac{18}{5}\right)\end{aligned}$$

7. **Ans (B)**



$$F = F_1 + F_2$$

$$F = P_{\text{com}} A$$

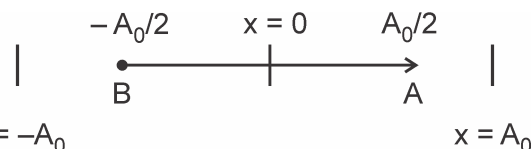
$$\begin{aligned}F_1 &= \rho g \left(\frac{2}{3} \times \frac{H}{2}\right) \times \frac{L}{2} \times \frac{H}{2} \\ &= \frac{\rho g L H^2}{12}\end{aligned}$$

$$F_2 = \left(\rho g \frac{3H}{4}\right) \frac{LH}{2} = \frac{3\rho g H^2 L}{8}$$

$$\begin{aligned}F &= \frac{\rho g L H^2}{12} + \frac{3\rho g L H^2}{8} \\ &= \frac{11\rho g L H^2}{24}\end{aligned}$$

8. **Ans (A)**

Initially



$$x = -A_0$$

$$A \text{ is at } \frac{A}{2} \text{ \& B is at } \frac{-A}{2}$$

$$x_A = A_0 \sin\left(\omega t + \frac{\pi}{6}\right)$$

$$x_A = A_0 \sin\left(\omega t - \frac{\pi}{6}\right)$$

$$\text{At } t \quad x_A = x_B$$

$$\sin\left(\omega t + \frac{\pi}{6}\right) = \sin\left(\omega t - \frac{\pi}{6}\right)$$

$$\sin \omega t \cos \frac{\pi}{6} + \cos \omega t \sin \frac{\pi}{6} = \sin \omega t \cos \frac{\pi}{6} - \cos \omega t \sin \frac{\pi}{6}$$

$$\Rightarrow \cos \omega t = 0$$

$$\omega t = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$v_A = -v_B \text{ so } \omega t = \frac{\pi}{2}$$

$$t = \frac{\pi}{2\omega} = \frac{\pi}{2\left(\frac{2\pi}{T}\right)} = \frac{T}{4}$$

9. **Ans (B)**

Normal is along $\hat{j}$ so y component of incident ray will be reversed.

$$n_{\text{reflected}} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$$

11. **Ans (C)**

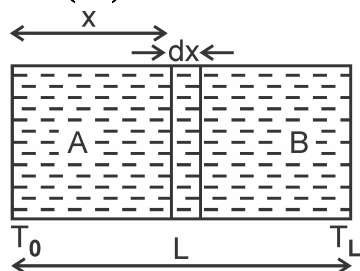
$$\frac{dQ}{dt} = -KA \frac{dT}{dx}$$

$$\frac{dQ}{dt} = -\frac{K_0 A}{a + bT} \frac{dT}{dx}$$

$$\frac{dQ}{dt} \int_0^1 dx = -K_0 A \int_{T_1}^{T_2} \frac{dT}{a + bT}$$

$$\frac{dQ}{dt} = \frac{AK_0}{b\ell} \ln \left[\frac{a + bT_1}{a + bT_2} \right]$$

12. **Ans (A)**

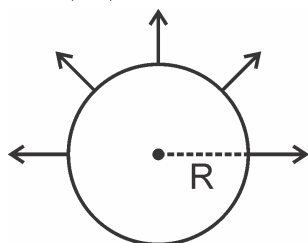


$$v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{5RT}{3M}}$$

$$dx = C \cdot dt = \sqrt{\frac{5R}{3M}} \left[T_0 + \left(\frac{T_L - T_0}{L} \right) x \right] dt$$

$$t = \frac{2L}{\sqrt{T_L} + \sqrt{T_0}} \sqrt{\frac{3M}{5R}}$$

13. **Ans (D)**



In equilibrium of upper half of loop

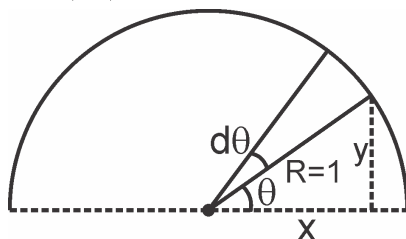
$$2kx = 4RS$$

$$x = \frac{2SR}{k}$$

$$2\pi R = \ell + \frac{2SR}{k}$$

$$R = \frac{k\ell}{2\pi k - 2S}$$

14. **Ans (A)**



$$x = \cos\theta$$

$$y = \sin\theta$$

$$dq = \lambda d\theta$$

$$= (4\cos^2\theta - 1) \sin\theta \lambda d\theta$$

$$= \lambda(4(1 - \sin^2\theta) - 1) \sin\theta d\theta$$

$$= \lambda(3\sin\theta - 4\sin^3\theta) d\theta$$

$$= \int \lambda \sin 3\theta d\theta$$

$$U = \lambda \left(\frac{-\cos 3\theta}{3} \right) \Big|_0^\pi = \frac{2}{3} \lambda$$

15. **Ans (A)**

Shifting of lens is

$$= 6 \times 1 = 6 \text{ cm}$$

$$u = -x$$

$$v = x + 6$$

$$\frac{x+6}{-x} = -2$$

$$\Rightarrow x = 6 \text{ cm}$$

$$u = -6 \text{ cm}$$

$$v = 12 \text{ cm}$$

$$\Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{12} + \frac{1}{6} = \frac{1}{f}$$

$$\Rightarrow f = 4 \text{ cm}$$

16. **Ans (C)**

$$\frac{hc}{\lambda} - \phi = e(5V_0) \quad \dots(1)$$

$$\frac{hc}{3\lambda} - \phi = e(V_0) \quad \dots(2)$$

Solving (1) & (2), we get

$$\phi = \frac{hc}{6\lambda}$$

17. **Ans (B)**

$$y' = \frac{d}{2} \text{ for a point exactly in front } S_1$$

$$\text{So, } \Delta x = \frac{yd}{D} + \frac{\frac{d^2}{2}d}{D}$$

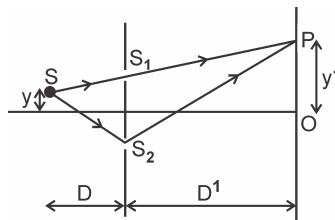
For maximum intensity $\Delta x = n\lambda$

$$\frac{yd}{D} + \frac{\frac{d^2}{2}d}{D} = n\lambda$$

$$\Rightarrow 0.5 \sin \pi t + \frac{(1)^2}{4} = n \times 500 \times 10^{-6} \times 10^3$$

$$\Rightarrow 0.5 \sin \pi t + 0.25 = 0.5n$$

$$\sin(\pi t) = \frac{0.5n - 0.25}{0.5}$$



For minimum 't' we have $n = 1$

$$\sin(\pi t) = \frac{0.5 - 0.25}{0.5} = 0.5$$

$$\pi t = \frac{\pi}{6}$$

$$t = \frac{1}{6} \text{ s}$$

18. Ans (C)

$$A = A_0 e^{-\frac{5t}{6}}$$

$$\Rightarrow \frac{A_0}{2} = A_0 e^{-\frac{5t}{6}}$$

$$\Rightarrow e^{-\frac{5t}{6}} = \frac{1}{2}$$

$$\therefore t = \frac{6}{5} \ln 2$$

19. Ans (D)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \rightarrow \text{differentiating twice w.r.t. time}$$

$$|a_i| = \frac{2m^3 u_0^2}{f}$$

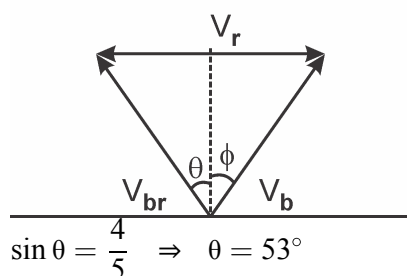
$u_0 \rightarrow$ volume of object
 $m \rightarrow$ magnification
 $f \rightarrow$ focal length.

20. Ans (B)

$$\text{Drift } X = (V_r - V_{br} \sin \theta) \frac{d}{V_{br} \sin \theta}$$

For minimum drift if $V_r > V_{br}$

$$\frac{dx}{d\theta} = 0 \Rightarrow \theta = \frac{V_{br}}{V_r}$$



(143° with the flow direction)

$$\text{Time} = \frac{36}{4 \cos 53^\circ} = \frac{36}{4(3/5)} = 15 \text{ sec.}$$

$$X_{\min} = \left[5 - 4 \left(\frac{4}{5} \right) \right] 15 = \frac{9}{5} \times 15 = 27 \text{m}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (6)

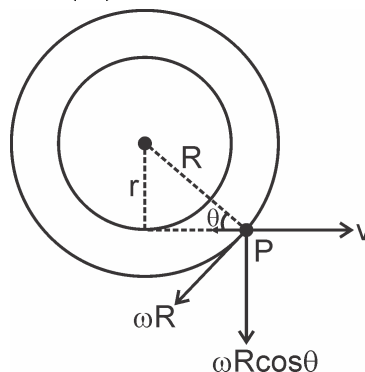
$$R_{\text{voltmeter}} = 6\Omega, R_{\text{ammeter}} = 0.5\Omega$$

$$R_{\text{eq}} = 10\Omega$$

$$I = \frac{30}{10} = 3 \text{A}$$

$$\text{Reading of voltmeter} = 1 \times 3 = 3 \text{ volt.}$$

2. Ans (3)



Horizontal velocity of point

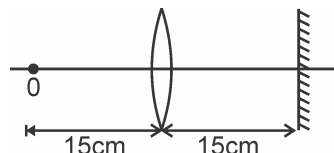
$$= v - \omega R \sin \theta$$

$$= \omega r - \omega R \left(\frac{r}{R} \right)$$

$$= 0$$

So velocity must be perpendicular.

3. Ans (60)



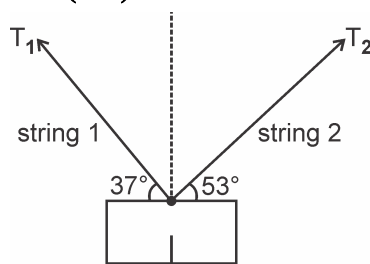
For lens

$$\frac{1}{V'} - \frac{1}{(-15)} = \frac{1}{30} \Rightarrow v' = 30 \text{cm}$$

So an image is formed 45cm behind the mirror which will act as an object for lens.

$$\frac{1}{V} - \frac{1}{(-60)} = \frac{1}{30} \Rightarrow v = 60 \text{cm}$$

4. Ans (14)



$$\frac{4T_1}{5} = \frac{3T_2}{5} \Rightarrow T_1 = \frac{3}{4} T_2$$

$$v = \sqrt{\frac{T}{\mu}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{3}{4}}$$

$$\ell_1 = 10 \times \frac{4}{5} = 8 \text{m}$$

$$\ell_2 = 10 \times \frac{3}{5} = 6 \text{m}$$

$$\Rightarrow \frac{t_1}{t_2} = \frac{\ell_1/v_1}{\ell_2/v_2} = \frac{8}{6} \sqrt{\frac{4}{3}} = \frac{3}{3\sqrt{3}}$$

5. Ans (6)



$$p = p_0 + \frac{2\gamma}{r} + h \rho g$$

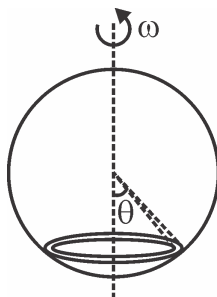
$$\Delta p = \frac{2\gamma}{r} + h \rho g$$

$$= \frac{2 \times 7 \times 10^{-3}}{0.35 \times 10^{-3}} + 2 \times 10^{-2} \times 10^4 = 240$$

$$\frac{\Delta p}{40} = 6$$

6. Ans (3)

When the shell rotates, current is induced due to motion of charge. To calculate magnetic induction at centre of the shell, rotating shell can be assumed to be composed of thin circular current carrying rings. Such a ring can be assumed as follows:
Consider a radius of the shell inclined at angle 'q' with the axis of rotation. This radius is rotated about the axis keeping q constant. Thus a circle is traced as shown in fig.



Its radius, $r = R \sin \theta$

Distance of its centre from centre of the shell,

$$x = R \cos \theta.$$

Now consider another radius inclined at angle $(\theta + d\theta)$. It is also rotated in the same way and another circle is traced. The portion between two circles forms a circular ring.

$$\text{Area of this ring} = 2\pi R d\theta = 2\pi R^2 \sin \theta d\theta$$

$$\text{Charge on this ring, } dq = s \cdot 2\pi R^2 \sin \theta d\theta$$

Since, angular velocity of the shell is ω , therefore, it completes $\frac{\omega}{2\pi}$ revolutions per second.

Hence, current associated with the ring considered,

$$i = \frac{2}{2\pi} dq = s \omega R^2 \sin \theta d\theta$$

Since, centre of the shell is a point lying on the axis of a circular coil of radius r , carrying current i at a distance x from centre of the coil, therefore, magnetic induction at centre of the shell due to this coil is

$$dB = \frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}}$$

$$= \frac{1}{2} \mu_0 s \omega R \sin^3 \theta d\theta$$

Hence, resultant magnetic induction at centre of the shell

$$B = \int dB = \frac{1}{2} \mu_0 s \omega R \int_0^\pi \sin^3 \theta d\theta = \frac{2}{3} \mu_0 s \omega R$$

7. Ans (120)

As here initially

$$P_I = P_0 = 1 \times 10^5 \text{ N/m}^2,$$

$$V_I = V_0 = 2.4 \times 10^{-3} \text{ m}^3,$$

$$T_I = T_0 = 300\text{K}$$

and finally

$$P_F = P_0 + \frac{Kx}{A} = 1 \times 10^5 + \frac{8000 \times 0.1}{8 \times 10^{-3}}$$

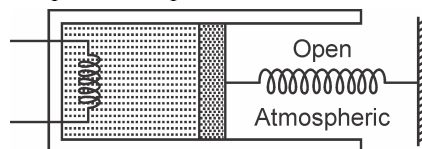
$$= 2 \times 10^5 \text{ N/m}^2,$$

$$V_F = V_0 + Ax = 2.4 \times 10^{-3} + 0.1 \times 8 \times 10^{-3}$$

$$= 3.2 \times 10^{-3} \text{ m}^3,$$

So from gas equation $PV = \mu RT$, i.e.

$$\frac{P_I V_I}{T_I} = \frac{P_F V_F}{T_F}$$



we have

$$T_F = \frac{P_F}{P_I} \times \frac{V_F}{V_I} \times T_I$$

$$= \frac{2 \times 10^5}{1 \times 10^5} \times \frac{3.2 \times 10^{-3}}{2.4 \times 10^{-3}} \times 300 = 800\text{K}$$

$$\text{As } DW = \int P dV,$$

$$\text{here } P = P_0 + (kx/A)$$

$$\text{So, } DW = \int_0^{0.1} \left(P_0 + \frac{kx}{A} \right) A dx$$

$$= \int_0^{0.1} (P_0 A + kx) dx$$

$$\text{i.e. } = P_0 Ax + \frac{1}{2} kx^2$$

$$= [10^5 \times 8 \times 10^{-3} \times 0.1 + \frac{1}{2} \times 8000 \times (0.1)^2]$$

$$\text{i.e. } DW = 80 + 40 = 120\text{J}$$

8. Ans (3)

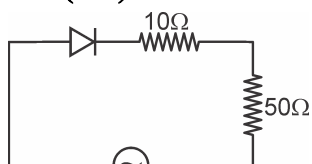
$$-1.51\text{eV} = \frac{-13.6\text{eV}}{n^2}$$

$$n^2 = \frac{13.6}{1.51} = 9$$

$$\therefore L = \frac{nh}{2\pi} = \frac{9h}{2\pi}$$

$$\Rightarrow x = 3$$

9. Ans (19)



$$I = \frac{20 - 0.7}{510} = 0.0378\text{A}$$

Peak voltage across load

$$= 0.378 \times 500$$

$$= 18.9\text{V}$$

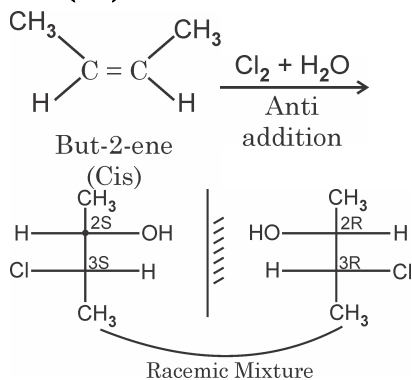
10. Ans (4)

displacement of top most point will be double of centre of the disk

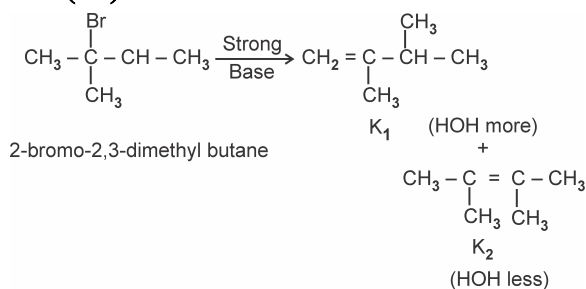
PART-2 : CHEMISTRY

SECTION-I

1. Ans (A)



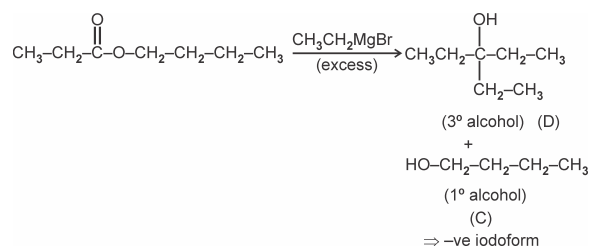
2. Ans (C)



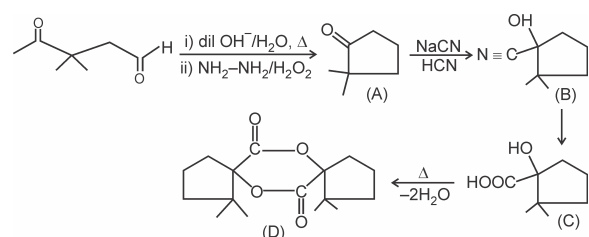
$\Rightarrow \text{K}_1 \text{ \& } \text{K}_2 \rightarrow \text{Positional isomer}$

$\Rightarrow \text{K}_2$ alkene will be formed by $\text{Et} - \text{O}^-$ (less steric crowded base)

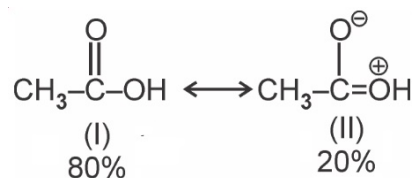
3. Ans (D)



4. Ans (D)



5. Ans (D)



$$\text{B.O} = 0.8 \times 2 + 0.2 \times 1 \quad \text{B.O.} = 0.8 \times 1 + 0.2 \times 2$$

$$= 1.8$$

$$= 1.2$$

$$A = 1.8$$

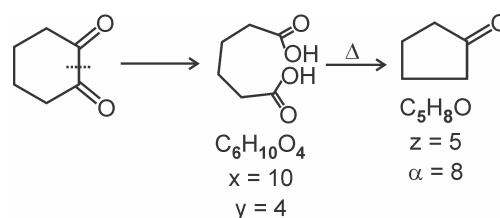
$$B = 1.2$$

$$A \times 5 = 1.8 \times 5 = 9.0$$

6. Ans (C)

NCERT Based

7. Ans (C)



$$(x - y + \alpha - 2z) = 10 - 4 + 8 - 2(5) = 4$$

8. **Ans (B)**

Energy observed for given photodissociation is equal to the sum of dissociation energy of A_2 and extra energy of excited atom.

Bond dissociation energy of A_2

$$= 200 \text{ K cal/mol}$$

$$= \frac{200 \times 10^3 \times 4.18}{6 \times 10^{23}}$$

$$= 13.9 \times 10^{-19} \text{ J/molecule}$$

The extra energy possessed by excited atom A^*

$$= 3 \times 1.6 \times 10^{-19} \text{ J}$$

$$= 4.8 \times 10^{-19} \text{ J}$$

$$E_{\text{absorbed}} = (13.9 + 4.8) \times 10^{-19} = 18.7 \times 10^{-19} \text{ J}$$

$$E = h\nu$$

$$18.7 \times 10^{-19} = 6.6 \times 10^{-34} \nu$$

$$\nu = 2.8 \times 10^{15} \text{ Hz}$$

9. **Ans (B)**

In allene, two CH_2 are perpendicular to each other



$\rightarrow$ (3) & (4) also correct

10. **Ans (D)**

for first bulb

$$(\text{Moles})_{O_2} = 1 \text{ mole} = (\text{Moles})_{SO_2}$$

After opening

Mass of SO_2 left in first bulb = 32g = 0.5 mole

Moles of SO_2 effused into second bulb = 1 – 0.5 = 0.5 mole

$$\frac{(\text{Moles})_{SO_2}}{(\text{Moles})_{O_2}} = \sqrt{\frac{32}{64}}$$

$$\frac{1}{2(\text{Moles})_{O_2}} = \frac{1}{\sqrt{2}}$$

$$(\text{Moles})_{O_2} = \frac{1}{\sqrt{2}} = \text{Moles of } O_2 \text{ in second bulb}$$

$$\text{weight of } O_2 \text{ in second bulb} = \frac{32}{\sqrt{2}} = 22.6 \text{ g}$$

11. **Ans (B)**

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{10 \times 2}{400} = \frac{5 \times 5}{T_2}$$

$$T_2 = 500 \text{ K}$$

$$q = C(T_2 - T_1) = 50(500 - 400)$$

$$= 5 \text{ KJ}$$

$$w = -1(V_2 - V_1) = -1(5 - 2) = -3 \text{ L atm}$$

$$= -0.3 \text{ KJ}$$

$$\Delta U = q + w$$

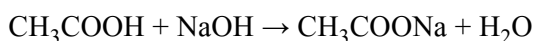
$$\Delta U = 5 - 0.3 = 4.7 \text{ KJ}$$

$$\Delta H = \Delta U + P_2 V_2 - P_1 V_1$$

$$= 4.7 + (5 \times 5 - 2 \times 10) \times 100 \times 10^{-3}$$

$$= 5.2 \text{ KJ}$$

12. **Ans (A)**



$$0.25 \quad 0.02$$

$$0.23 \quad - \quad 0.02$$

$$pH = pK_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

$$= pK_a + \log \left(\frac{0.27}{0.23} \right)$$

$$[H^+] = K_a \times \frac{0.23}{0.27} = 2 \times 10^{-5} \times \frac{0.23}{0.27}$$

$$= 1.7 \times 10^{-5} \text{ M}$$

$$[OH^-] = \frac{K_w}{[H^+]} = 5.9 \times 10^{-10} \text{ M}$$

13. **Ans (A)**

$$\Delta T_f = i K_f m$$

$$i = \frac{\Delta T_f}{K_f m} = \frac{0.012}{2 \times 0.001} = 6$$

for 100% dissociation, $i = n$

Hence, 6 ions will be in the solution

14. **Ans (B)**

$$Lu^{+3} = 4f^{14}$$

$$Tb^{+2} = 4f^9$$

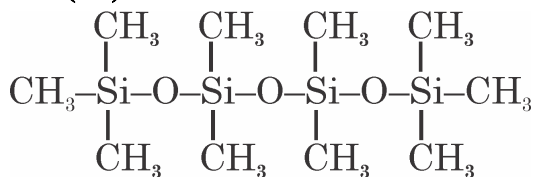
$$Ce^{4+} = 4f^0$$

$$Pm^{+2} = 4f^5$$

15. **Ans (D)**

F^- is a weak field ligand, iii and iv are outer orbital complex.

16. Ans (D)



17. Ans (B)

When purified Al_2O_3 is mixed with NaAlF_6 or CaF_2 , Melting point of the mixture decreases and brings conductivity.

18. **Ans (B)**

Final volume of gas at 608 torr

$$V_2 = \frac{P_1 V_1}{P_2} = \frac{760 \times 1}{608}$$

$$V_2 = 1250 \text{ ml}$$

$$\text{Volume occupied by gas} = 1000 - \frac{16}{1.6} = 990 \text{ ml}$$

$\therefore$ Volume of gas adsorbed by charcoal

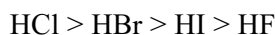
$$= 1250 - 990 = 260 \text{ ml}$$

Volume of the gas adsorbed per gram of charcoal

$$= \frac{260}{16} = 16.25 \text{ ml/g}$$

19. **Ans (D)**

→ Volatile nature order of halogen acids



→ NH_3 has greater forces of attraction as compared to HCl .

$$\rightarrow \text{B.P} \Rightarrow \text{H}_2\text{O} > \text{HF}$$

→ $K_{a1} \Rightarrow$ maleic acid $>$ fumaric acid

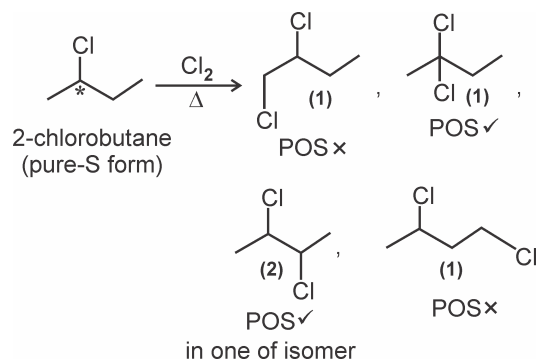
20. Ans (B)

2nd is correct answer

PART-2 : CHEMISTRY

SECTION-II

1. Ans (10)

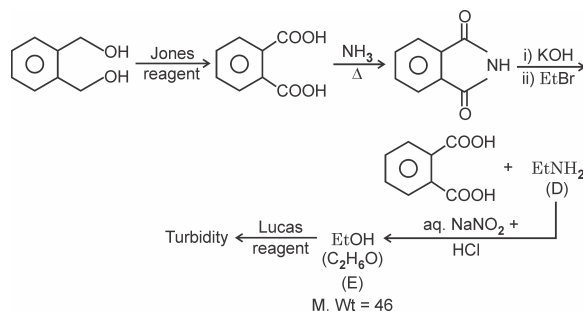


$$x = 5$$

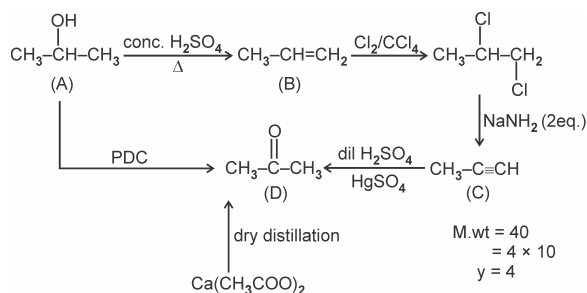
$$y = 2$$

$$\mathbf{x} \times \mathbf{y} = 5 \times 2 = 10$$

2. Ans (46)



3. Ans (4)



4. Ans (4)

$\lambda_{\text{eq}}^{\infty}, \text{CH}_3\text{COOH}$

$$= \lambda_{\text{eq}}^{\infty}, \text{CH}_3\text{COONa} + \lambda_{\text{eq}}^{\infty}, \text{HCl} - \lambda_{\text{eq}}^{\infty} \text{NaCl}$$

$$= 91 + 426.16 - 126.45$$

$$= 390.7$$

$$\alpha = \frac{\lambda_{\text{eq}}}{\lambda_{\text{eq}}^{\infty}}$$

$$\lambda_{\text{eq}} = 0.01 \times 390.7 \simeq 4 \text{ Scm}^2\text{eq}^{-1}$$

5. **Ans (4)**

$$r = -\frac{d[A]}{dt} = -\frac{-d[B]}{dt} = +\frac{d[C]}{dt}$$

$$= K[A]^{\alpha}[B]^{\beta}[C]^{\gamma}$$

$$\text{Assume } -\frac{d[B]}{dt} = x$$

$$x = K[A]^{\alpha}[B]^{\beta}[C]^{\gamma}$$

$$4x = K[A]^{\alpha}[\sqrt{2}B]^{\beta}[C]^{\gamma}$$

$$\therefore \beta = 4$$

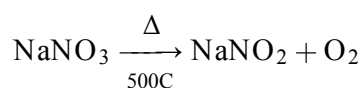
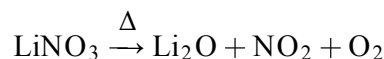
6. **Ans (7)**

meq of produced NH_3 = meq of consumed H_2SO_4

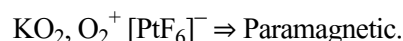
$$\text{percentage of N} = \frac{14 \times 12.75 \times 2 \times 100}{1000 \times 0.85}$$

$$= 42\%$$

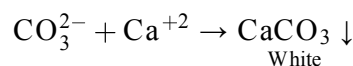
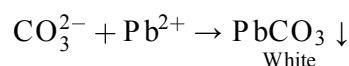
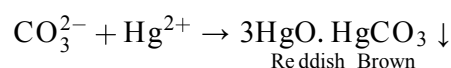
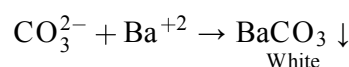
7. **Ans (4)**



8. **Ans (4)**



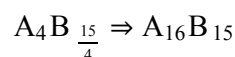
9. **Ans (3)**



10. **Ans (1)**

$$A = 4$$

$$B = 4 - \frac{1}{4} = \frac{15}{4}$$



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

$$\int (e^{2x} + 2e^x - e^{-x} - 1) \cdot e^{(e^x + e^{-x})} dx$$

$$\text{Let } e^x = t, dx = dt/t$$

$$= \int \left(t^2 + 2t - \frac{1}{t} - 1 \right) e^{(t + \frac{1}{t})} \frac{dt}{t}$$

$$= \int \left[\frac{(t^2 - 1)(t + 1)}{t^2} + 1 \right] e^{(t + \frac{1}{t})} dt$$

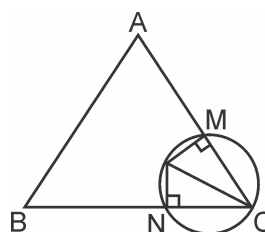
$$= \int (t + 1) \left(1 - \frac{1}{t^2} \right) e^{(t + \frac{1}{t})} dt + \int e^{(t + \frac{1}{t})} dt$$

$$= (e^x + 1)e^{(e^x + e^{-x})} + c$$

$$g(x) = 1 + e^x$$

$$g(0) = 2$$

2. **Ans (A)**



As PMCN is a cyclic quadrilateral

hence $\angle PCM = \angle PNM$

3. **Ans (D)**

$$S = \sum_{r=1}^{2n} ((1-x)^n)^{r/n} = \sum_{r=1}^{2n} (1-x)^r$$

$$= \frac{(1-x)[1 - (1-x)^{2n}]}{1 - (1-x)}$$

$$S = \frac{1-x}{x} - \frac{(1-x)^{2n+1}}{x}$$

Coefficient of x^{2n} in S

$$= \text{Coefficient of } x^{2n+1} \text{ in } [(1-x) - (1-x)^{2n+1}]$$

$$= -(-^{2n+1}C_{2n+1}) = 1$$

4. **Ans (A)**

$$f(3-) \geq f(3)$$

$$m^2 - 9m - 9 \geq 1$$

$$m \in (-\infty, -1] \cup [10, \infty)$$

5. **Ans (D)**

$$x^2 - 3x + 2 \geq 0$$

$$\Rightarrow x \leq 1 \text{ or } x \geq 2$$

$$4x - x^2 - 3 \geq 0$$

$$\Rightarrow x^2 - 4x + 3 \leq 0$$

$$\Rightarrow 1 \leq x \leq 3$$

i.e., either $x = 1$ or $2 \leq x \leq 3$

But $x = 1$ does not satisfy the equation given

$$\text{also } 0 < \cos^{-1} \sqrt{4x - x^2 - 3} \leq \frac{\pi}{2}.$$

So, $0 < \text{L.H.S.} < \pi$

i.e., the given equation has no solution.

6. **Ans (A)**

$$\sim [(p \rightarrow q) \vee (p \vee q)]$$

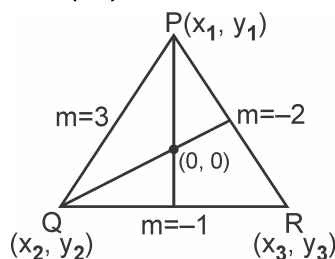
$$\sim (p \rightarrow q) \wedge \sim (p \vee q)$$

$$(p \wedge \sim q) \wedge (\sim p \wedge \sim q)$$

$$(p \wedge \sim p) \wedge \sim q$$

$$c \wedge \sim q = c$$

7. **Ans (C)**



$$\frac{y_1}{x_1} = 1, \quad \frac{y_2}{x_2} = \frac{1}{2}$$

$$\frac{y_3}{x_3} = -\frac{1}{3}$$

Slopes of QR, RP and PQ are $-1, -2, 3$.

$$m_{QR} = \frac{\frac{x_2}{2} + \frac{x_3}{3}}{x_2 - x_3} = -1 \Rightarrow 9x_2 = 4x_3$$

$$\text{Similarly } 9x_1 = 5x_3$$

$$G = (h, k) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{x_1 + \frac{x_2}{2} - \frac{x_3}{3}}{3} \right)$$

$$\Rightarrow \frac{k}{h} = \frac{2}{9}$$

$$9y = 2x$$

8. **Ans (C)**

$$y = x^2 \text{ and } y = 2x$$

$$x^2 = 2x$$

$$x = 2$$

$$x > 2 \text{ or } x < -2$$

9. **Ans (D)**

$$\begin{aligned} & \sum_{\alpha=1}^{\infty} \sin^{-1} \left(\frac{\sqrt{\alpha^2 + 2\alpha} - \sqrt{\alpha^2 - 1}}{\alpha(\alpha+1)} \right) \\ &= \sum_{\alpha=1}^{\infty} \sin^{-1} \left(\frac{\sqrt{(\alpha+1)^2 - 1} - \sqrt{\alpha^2 - 1}}{\alpha(\alpha+1)} \right) \\ &= \sum_{\alpha=1}^{\infty} \sin^{-1} \left[\frac{1}{\alpha} \sqrt{1 - \frac{1}{(\alpha+1)^2}} - \frac{1}{(\alpha+1)} \sqrt{1 - \frac{1}{\alpha^2}} \right] \\ &= \sum_{\alpha=1}^{\infty} \left(\sin^{-1} \frac{1}{\alpha} - \sin^{-1} \frac{1}{\alpha+1} \right) \\ &= \sin^{-1} 1 - \sin^{-1} \frac{1}{2} + \sin^{-1} \frac{1}{2} - \sin^{-1} \frac{1}{3} + \dots \\ &+ \sin^{-1} \frac{1}{n} + \sin^{-1} \frac{1}{n+1} \\ &S_{\infty} = \frac{\pi}{2} \end{aligned}$$

10. **Ans (C)**

$$|\vec{a}\vec{b}\vec{c}| = 30$$

$$|abc \sin \theta \cos \phi| = 30 \Rightarrow$$

$\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular

$$\begin{aligned} k &= (2\vec{a} + \vec{b} + \vec{c}) \cdot [(\vec{a} \times \vec{c}) \times (\vec{a} - \vec{c}) + \vec{b}] \\ &= 2a^2c^2 + b^2 + a^2c^2 = 3a^2c^2 + b^2 \\ &= 309 \end{aligned}$$

$$\text{Then, } \left(\frac{k}{103} \right) = 3$$

12. **Ans (C)**

We have

$$\int_0^{\pi/2} (\sin x + t \cos x) dx = 1 + t$$

$$I = \int_0^{\pi/2} (\sin x + t \cos x)^2 dx$$

$$= \frac{\pi}{4} t^2 + t + \frac{\pi}{4}$$

$$g(t) = \frac{t+1}{\sqrt{at^2+t+a}} \quad \alpha = \frac{\pi}{4}$$

$$g'(t) = \sqrt{at^2+t+a} - \frac{t+1}{2\sqrt{at^2+t+a}} (2at+1) = 0$$

$$(2at^2 + 2t + 2a) - (2at^2 + t + 2at + 1) = 0$$

$$t + 2a - 2at - 1$$

$$(t-1)(1-2a) = 0$$

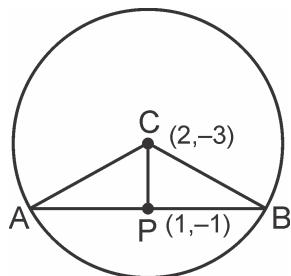
$$t = 1$$

13. **Ans (B)**

$$S = x^2 + y^2 - 4x + 6y - 3 = 0$$

$$C(2, -3)$$

$$r = \sqrt{4 + 9 + 3} = 4$$



Length of chord is shortest when CP is $\perp$ to chord.

$$AB = 2\sqrt{r^2 - cp^2}$$

$$= 2\sqrt{16 - 5} = 2\sqrt{11}$$

14. **Ans (C)**

$$f(x) \times \frac{dy}{dx} = f^2(x) + f(x)y + f'(x)y$$

$$\frac{dy}{dx} - \left(1 + \frac{f'(x)}{f(x)}\right)y = f(x)$$

$$I.F. = e^{\int -\left(1 + \frac{f'(x)}{f(x)}\right) dx}$$

$$= e^{-x - \ln f(x)} = \frac{e^{-x}}{f(x)}$$

$$\frac{e^{-x}}{f(x)} y = \int \frac{e^{-x}}{f(x)} \times f(x) dx$$

$$\frac{e^{-x} y}{f(x)} = -e^{-x} + c$$

$$y = -f(x) + ce^x f(x)$$

15. **Ans (B)**

Equation of tangent to the parabola $y = 4x$

$y = mx + \frac{1}{m}$ which is also tangent to the circle

$$p = r.$$

$$\frac{|-2m + \frac{1}{m}|}{\sqrt{1 + m^2}} = 1$$

$$|1 - 2m^2| = m\sqrt{1 + m^2}$$

$$3m^4 - 5m^2 + 1 = 0$$

$$\sum_{i=1}^4 m_i^2 = 4 \times \frac{5}{6} = \frac{10}{3}$$

16. **Ans (B)**

$$\int_0^{10} [x]^3 \{x\} = \int_0^1 0 dx + \int_1^2 \{x\} dx + \int_2^3 2^3 \{x\} dx + \dots$$

$$\dots + \int_9^{10} 9^3 \{x\} dx$$

$$= (1^3 + 2^3 + 3^3 + \dots + 9^3) \int_0^1 \{x\} dx$$

$$= \frac{2025}{2}$$

17. **Ans (B)**

$$P(A) = \frac{5}{10}; \quad P(B) = \frac{3}{10}; \quad P(C) = \frac{2}{10}$$

Due to accident

$$P(A) = \frac{1}{3}; \quad P(B \cup C) = \frac{2}{3}$$

$$P(B) = \frac{3}{5} \cdot \frac{2}{3} = \frac{2}{5}$$

$$P(C) = \frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$$

18. **Ans (B)**

$$\alpha(3\alpha^2 - 5\alpha + 7) - 1, \beta(3\beta^2 - 5\beta + 7) + 2$$

$$-1, 2$$

Sum of root's = 1

Product = -2

$$x^2 - x - 2 = 0$$

19. **Ans (A)**

$$-4 + i \leq f(i) \leq i + 4$$

$$-3 \leq f(1) \leq 5 \quad {}^3C_1$$

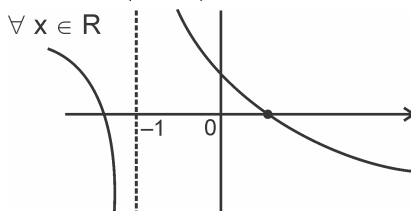
$$-2 \leq f(2) \leq 6 \quad {}^3C_1$$

Similarly $f(3), f(4)$ and $f(5)$

So, total such function $({}^3C_1)^5$

20. **Ans (D)**

$$f'(x) = \frac{-5}{(x+1)^6} - 7 + 2 \cos x < 0$$



PART-3 : MATHEMATICS

SECTION-II

1. **Ans (1)**

$$f(a) = 0 \text{ and } f(b) = 0$$

Then atleast one root of $f'(x)$ lying in (a, b) .

2. **Ans (3)**

Clearly the triangle is an equilateral triangle such that $AB = BC = CA = \sqrt{2}$.

So distance of centre 'O' from the sides is equal to inradius of the triangle.

$$\text{So, } \ell_1 = \ell_2 = \ell_3 = \frac{\Delta}{s} = \frac{\frac{\sqrt{3}}{4} \times (\sqrt{2})^2}{\frac{3}{2}\sqrt{2}} = \frac{1}{\sqrt{6}}$$

$$\text{So, } \sqrt{6}(\ell_1 + \ell_2 + \ell_3) = \sqrt{6} \times \frac{3}{\sqrt{6}} = 3$$

3. **Ans (5)**

Equation of the plane passing through the intersection of P_1 and P_2 is

$$(3x - y + 2z - 4) + \lambda(x + y + z - 2) = 0$$

it passes through (2, 2, 1).

$$\text{Hence } \lambda = \frac{-2}{3} \text{ required plane is}$$

$$\frac{7x}{2} - \frac{5y}{2} + 2z = 4$$

$$a + b + c = 5$$

4. **Ans (5)**

$$(6 \tan 2\theta - 12) \tan \theta = 0$$

$$\tan \theta = 0, \tan 2\theta = \frac{12}{5}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{12}{5}$$

$$6 \tan^2 \theta + 5 \tan \theta - 6 = 0$$

$$\tan \theta = \frac{2}{3}, \frac{-3}{2}$$

$$3 \sin^2 \theta - 2 \sin \theta \cos \theta = 0$$

$$\sin \theta (3 \sin \theta - \cos \theta) = 0$$

$$\sin \theta = 0, \tan \theta = \frac{2}{3}$$

$$\text{Common solution } \sin \theta = 0 \text{ and } \tan \theta = \frac{2}{3}$$

Number of solutions = 5.

5. **Ans (20)**

Centre of circle = (4, -5)

$$\text{Radius} = \sqrt{|-4 + 5i|^2 - (-40)} = 9$$

$$a = \max |z - (-2 + 3i)| = 9 + 10 = 19$$

$$b = \min |z - (-2 + 3i)| = 10 - 9 = 1$$

$$a + b = 20$$

6. **Ans (85)**

$${}^{11}C_2 + 9 \times {}^{11}C_1 + 17 \times {}^{11}C_0 = 171$$

$$\left[\frac{171}{2} \right] = 85$$

7. **Ans (16)**

$$\tan 2\theta = \frac{2h}{a-b} = \frac{2}{5-3}$$

$$\tan 2\theta = 1$$

$$2\theta = \frac{\pi}{4}$$

$$\theta = \frac{\pi}{8}$$

8. **Ans (2)**

$$x^2 + \log_2(x^2 + 2^x) = 2^x + x + 1$$

$$x^2 + \log_2(x^2 + 2^x) = 2^x + \log_2 2^{x+1}$$

$$x^2 + \log_2(x^2 + 2^x) = 2^x + \log_2(2^x + 2^x)$$

Solution only when $x^2 = 2^x$

$x = 2, 4$ only the solution.

9. **Ans (1)**

$$T_n = \frac{H_{n+1} - H_n}{H_n H_{n+1}}$$

$$\text{As } H_{n+1} - H_n = \frac{1}{n+1}$$

$$T_n = \frac{1}{H_n} - \frac{1}{H_{n+1}}$$

$$T_1 + T_2 + T_3 + \dots + T_n$$

$$= \left(\frac{1}{H_1} - \frac{1}{H_2} \right) + \left(\frac{1}{H_2} - \frac{1}{H_3} \right) + \dots + \left(\frac{1}{H_n} - \frac{1}{H_{n+1}} \right)$$

As $n \rightarrow \infty$

$$T_1 + T_2 + T_3 + \dots + 0 = 1.$$

10. **Ans (31)**

$$\frac{(x-3)^{-\frac{|x|}{x}} \sqrt{(x-4)^2(17-x)}}{\sqrt{-x}(-x^2+x-1)(|x|-32)} < 0$$

For $\sqrt{-x}$ to be defined $x < 0$

Since, $\sqrt{(x-4)^2}$, $\sqrt{-x}$ are +ive and $-x^2+x-1$ is always negative.

$$\frac{(x-3)(x-17)}{x+32} > 0 \quad x < 0$$

$$\frac{1}{x+32} > 0 \Rightarrow -32 < x < 0$$

$$\begin{array}{ccccccc} & - & & + & & - & & + \\ & -32 & & 3 & & 17 & & \end{array}$$

Number of such integers = 31.